

What is solidification?

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Solidification is a transformation of liquid into solid below freezing point. Solidification is a comprehensive process of transformation of the melt metals and alloys into a solid piece, involving formation of dendrites, segregation which involves change in composition, zone formation in final structure of the casting and microporosity formation during shrinkage.

* The steps through which metals are solidified -

1. The formation of stable nuclei in the melt (nucleation)
2. The growth of new nuclei into crystals.

* **Nuclei** - A cluster of atoms which are bonded together larger than the critical size is called nuclei.

* **Embryo** - A cluster of atoms which are bonded together less than the critical size is called embryo.

Critical Radius - critical radius is the minimum particle size from which an aggregate is thermodynamically stable.

Or,

Critical Radius is the lowest radius ~~formed~~ formed by the atoms or molecules clustering together before a new phase inclusion is viable and begins to grow/ The dimension defining the boundary

between an embryo and nucleus is the critical radius.

* **Grain** - The crystals of solidified metal are called grain.
or, A single crystal in a polycrystalline aggregate.

* **Difference between embryo and nucleus**

* Nucleus is larger than the critical size, embryo is less than the critical size.

* Nucleus is stable and it can grow into a crystal, embryo are continuously being formed and redissolved in the molten metal due to the agitation of the atoms.

* **The criteria to take place heterogenous nucleation-**

1. The solid nucleating agent must be wetted by the liquid metal.

2. The liquid should solidify on the nucleating agent easily, for which their formation of structure should be same.

* **Alloy** - A mixture of two or more metals or a metal and a non-metal.

Solidification: Solidification is a phase transition process, where liquid turns into solid under freezing point.

☐ The steps through which metals are solidified—

1. The formation of stable nuclei in the melt called nucleation.
2. The growth of nuclei into crystals and the formation of a grain structure.

☐ There are two types of nucleation

- (i) Homogenous
- (ii) Heterogenous

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Homogenous Nucleation: In a liquid melt when the metal itself provides atoms to form a nucleus is called homogenous Nucleation.

Heterogenous Nucleation: The formation of very small regions of a solid phase at the interfaces of solid impurities is called heterogenous nucleation.

It is a nucleation that occurs in a liquid on the surfaces of its container, insoluble impurities and other structural materials that lower the critical size free energy required to form a stable nucleus.

☐ Heterogenous nucleation takes place on the nucleating agent because the surface free energy to form a stable nucleus is lower on this material than the pure liquid itself. Since the surface free energy is lower for heterogenous nucleation, the total free energy change

For the formation of a stable nuclei nucleus will be lower and the critical size of the nucleus will be ^{bigger} ~~smaller~~. Thus, a much smaller amount of under-cooling is required to form a stable nucleus produced by heterogenous nucleation.

Derivative the equation : $r^* = \frac{-2\gamma}{\Delta G_v}$

When a pure liquid metal such as lead is cooled below its equilibrium freezing temperature, the driving energy for the liquid to solid transformation is the difference in the volume free energy G_v of the liquid and that of solid. If G_v is the change in free energy change for a spherical nucleus of radius 'r' is $\frac{4}{3}\pi r^3 G_v$. Since the volume of a sphere is $\frac{4}{3}\pi r^3$. There is an opposing energy to the formation of embryo and nuclei. The energy required to form the surface of these particles. The energy needed to create a surface for those spherical particles, ΔG_s is equal to the specific surface energy of the particle γ . The surface free energy of the area $4\pi r^2 \gamma$ where $4\pi r^2$ is the surface area of a sphere. The total free energy associated with the formation of an embryo or nucleus, which is the sum of the volume free energy and surface free energy changes. In equation form, the total free energy change for the formation of a spherical embryo or nucleus of radius r formed in a freezing pure metal is,

$$\Delta G_T = \frac{4}{3}\pi n^3 \Delta G_V + 4\pi n^2 \gamma = \Delta G_V + \Delta G_S$$

Where,

ΔG_T = total free energy change

n = radius of embryo or nucleus

ΔG_V = volume free energy

γ = specific surface free energy

A relationship among the size of the critical nucleus, surface free energy and volume free energy for the solidification of a pure metal can be obtained by differentiating. The differential of the total free energy ΔG_T with respect to ' n ' is zero when $n = n^*$ since the total free energy versus radius of the embryo or nucleus plot is then at a maximum and the slope $d(\Delta G_T)/dn = 0$. Thus,

$$\Delta G_T = \frac{4}{3}\pi n^3 \Delta G_V + 4\pi n^2 \gamma$$

$$\Rightarrow \frac{d(\Delta G_T)}{dn} = \frac{d}{dn} \left(\frac{4}{3}\pi n^3 \Delta G_V + 4\pi n^2 \gamma \right)$$

$$\Rightarrow 0 = \frac{4}{3}\pi \Delta G_V \times 3n^2 + 4\pi \times 2n \times \gamma$$

$$\Rightarrow 0 = 4\pi n^2 \Delta G_V + 8\pi n \gamma$$

$$\Rightarrow 4\pi n^2 \Delta G_V = -8\pi n \gamma$$

$$\Rightarrow n^* = \frac{-8\pi \gamma}{4\pi \Delta G_V} = \frac{-2\gamma}{\Delta G_V}$$

(Showed)

Grain: Crystals in solidified metals are called grain.

Grain boundaries: The surfaces between grains are called grain boundaries.

□ Grain are two types -

(i) Equiaxed, (ii) Columnar

Equiaxed grain - The crystals can grow about equally in all direction. They can find adjacent to a cold mold wall.

Condition: Large amounts of undercooling near the wall creates a relatively high concentration.

Columnar grain: The crystal grow perpendicular to the mold faces since large thermal gradients were present in those directions. Here a metal solidifies slowly in the presents of a steep temperature.

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* Types of solutions

1. Liquid-liquid solution
2. Solid-liquid solution
3. Solid-solid solution

Q What is solid solution? Mention the types of solid solution?

Solid solution is an alloy where one or more solute atoms disappeared in a single phase solvent without exiting the phase of solutes.

Types of solid solution

In general there are two types of solid solution.

1. Substitutional solid solution
2. Interstitial solid solution

There are also two types of solid solution in arrangement of the atom.

1. Order solid solution
2. disorder solid solution

Substitutional solid solution: A solid solution in which solute atoms replace some solvent atoms so that they lie at normal atom sites of the crystal structure are called as substitutional solid solution.

Substitutional solid solution is formed by two elements. Solute atoms can be substitute for parent solvent atoms in a crystal

lattice lattice

Interstitial solid solution: A solid solution in which solute atoms are located in the interstices or voids of the solvent lattice is called interstitial solid solution.

In interstitial solid solution the solute atoms fit into the space between the solvent or parent atoms. This space are called interstices. Interstitial solid solution can form when one atom is much smaller than another.

Example of atoms that can form interstitial solid solution due to their small size are carbon, hydrogen, nitrogen, oxygen etc.

□ The condition of substitutional solid solution -

There are some conditions of substitutional solid solution which are given below -

1. atomic size must be within 15%
2. crystal structure must be same.
3. The two element should have the same valence.
4. There should be no appreciable differences in the electronegativities of the two elements.

☐ Write down the type of solid solidification by the basis of arrangements of liquid.

1. Order solid solution: When the solute will situate by orderly in interstitial space of an atom on the solvent in which is called the order solid solution.

2. Disorder solid solution: When the solute will situate at random in interstitial space of an atom on the solvent in which is called disorder solid solution.

☐ Relation between under cooling, grain size and strength.

1. Under cooling $\uparrow$

critical radius $\downarrow$

grain growing size $\downarrow$

The number of grain $\uparrow$

(ধূস্র দ্রুত cool করা হলে ছোট ছোট grain তৈরি হবে
বীড়ে বীড়ে cool করা হলে বড় বড় grain তৈরি হবে।)

2. Under cooling $\downarrow$

critical radius $\uparrow$

grain growing size $\uparrow$

The number of grain $\downarrow$

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Q How under cooling impact on critical radius size?

We know,

$$r^* = \frac{2\gamma T_m}{\Delta H_f \Delta T} \text{ where}$$

$$r^* \propto \frac{1}{\Delta T}$$

r^* = critical radius of nucleus

γ = surface free energy

ΔH_f = latent heat of fusion

T_m = Melting temperature

ΔT = Under cooling temperature.

*

Under cooling temperature $\propto$ volume free energy

Under cooling temperature doesn't have any impact on surface free energy.

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